

Fuzzifying the measured dirtiness—the degree of membership is where the value intersects the Dirty and Filthy sets

0.4 Dirty and 0.6 Filthy. The numbers are called *degrees of membership*, and mathematically represent the value's position in the sets.

To find out how much time it's going to take to wash these clothes, we need to see what happens when we apply our rules to this value of dirtiness.

I Want The Truth!

Another new term we come across here is *degree of truth*—a mathematical representation of the extent to which our measured value satisfies a rule. Let's look at our conditions one by one, and consequently how much truth the corresponding results have.

clothes are Clean

They're not, so the result *time = 10 minutes* has no truth

clothes are Dirty

They're 0.4 Dirty, so the result *time = 45 minutes* has a 0.4 degree of truth

clothes are Filthy

They're 0.6 Filthy, so the result *time = 90 minutes* has a 0.6 degree of truth

Now that we have this information, we need to “de-fuzzify” it to get a single, non-fuzzy output.

Making Sense Of It All

We know that if the clothes were perfectly dirty (a degree of truth of 1 for the second rule, 0 for the others), it would take 45 minutes to wash them. Similarly, it's 10 minutes for perfectly clean clothes and 90 for perfectly filthy clothes. If we plot these values on a graph, we get something like this:

The Fuzz-less Life

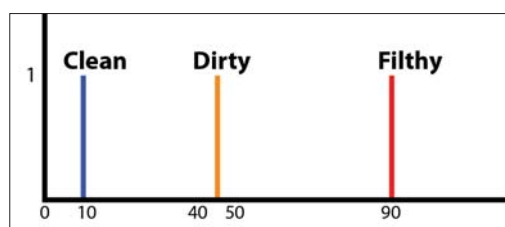
What would you do if you had to design a simple, smart washing machine (how's that for a mind-bender?) like the one we spoke about? One catch, though—pretend that fuzzy logic doesn't exist.

You'd need to develop a dirt sensor that's dead accurate even in a tangle of clothes, of course—but that's not the worst part. You'd need to develop a mathematical relationship between the dirtiness of clothes and the time that it would take to clean them. Even if you could, it'd end up being a complicated quadratic equation in the eighth degree, with fractional coefficients to boot (or as some mathematicians would say, “mind-numbing pain”). Just for fun, what if you were asked to factor in the concentration of the detergent used? Or water temperature?

And then, what if you (miraculously) did it all, only to find out that all the seemingly right calculations in the lab fell apart in the real world?

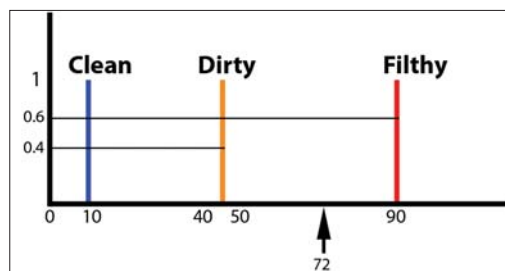
Fuzzy logic saves lives, you'll agree.

Remember that fuzzy logic systems aren't supposed to be overly complex—too many variables lead to too many rules



The relationship between time and degree of truth

Now let's put our findings—viz. 0.4 Dirty and 0.6 Filthy—on this graph. Using a mathematical technique called the Centre of Gravity method (which we won't get into here), we find that the answer we want is 72 minutes.



Voila! An answer!

Complicating Matters

We've just designed a very simple washing machine here, but you can see where all this can lead: you can bring more variables into the picture—clothes' greasiness, water temperature, the weight of the load (to optimally utilise water), and so on. It can be quite the intellectual exercise, this.

Just remember that fuzzy logic systems aren't supposed to be overly complex—too many variables lead to too many rules (just adding the “greasiness” variable to the machine above causes the number to increase to six), resulting in a lot of inefficiency.

Don't forget, the washing machine isn't the only place you'll find fuzzy logic used. It's been used in digital cameras to isolate faces from the rest of the scene, optical drives and heaven knows where potential applications lie.

Big Picture

Fuzzy logic is, as you might have guessed, the first step to artificial intelligence. After all, it is trying to duplicate the human thought process, no matter how primitively. For all its complexity, even game AI has its roots in a fuzzy logic system (“If health is low, take cover”—sound about right?).

Unlike true AI (which we've talked about a lot in these pages), though, fuzzy logic systems are perfectly realistic, and we'll continue to see them grow in applications where you don't need the mechanical versions of Einstein. We see them perfecting speech recognition systems, perhaps even bringing emotion to text-to-speech, and when the nanobots come, it'll be fuzzy logic telling them the difference between the harmful and harmless cells in your body. ■

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